

Luoye Chen

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Department of Agricultural and Consumer Economics

College of ACES, University of Illinois at Urbana Champaign, Urbana, IL 61801

EDUCATION

University of Illinois at Urbana-Champaign	<i>Ph.D. in Applied Economics</i>	<i>Expected: May 2022</i>
Peking University	<i>Master in Economics</i>	<i>2016</i>
Peking University	<i>Bachelor in Management</i>	<i>2014</i>
Peking University	<i>Bachelor in Economics (Minor)</i>	<i>2014</i>

WORK IN PROGRESS

- “The Impact of Prevented Planting Policy on Farming Practices”.
- “Vanishing Birds: the Impact of Grassland-cropland Conversions”.

JOB MARKET PAPER

- “Weather Shocks and Economic Triggers of Cropland Change in the US: A Fine-scale Spatial Analysis” (with Madhu Khanna).

WORKING PAPERS

- Luoye Chen and Madhu Khanna, “Heterogeneous Effects of a Changing Climate on Bird Biodiversity and Adaptability” (*Under Review at Nature Sustainability, lead author*).
- Xiaoguang Chen, Luoye Chen, Wei Xie, Nathan Mueller, Steven Davis, “Air Pollution and Flight Delays in China” *Revise and Resubmit, Journal of Environmental Economics and Management*. (first authorship is jointly shared by Xiaoguang Chen, Luoye Chen, and Wei Xie)

PEER-REVIEWED PUBLICATIONS

- **Luoye Chen**, Elena Blanc-Betes, Tara W. Hudiburg, Daniel Hellerstein, Steven Wallander, Evan H. DeLucia, Madhu Khanna, 2021. “Assessing the Returns to Land and Greenhouse Gas Savings from Producing Energy Crops on Conservation Reserve Program Land.” *Environmental Science & Technology*. (IF: 9.028, lead author) [LINK]
- **Luoye Chen**, Deepayan Debnath, Jia Zhong, Kelsie Ferin, Andy VanLooke, and Madhu Khanna, 2021. “The Economic and Environmental Costs and Benefits of the Renewable Fuel Standard.” *Environmental Research Letters*. (IF: 6.793, lead author) [LINK]
- Kelsie M. Ferin, **Luoye Chen**, Jia Zhong, Sarah Acquah, Emily A. Heaton, Madhu Khanna, and Andy VanLooke, 2021. “Water Quality Effects of Economically Viable Land Use Change in the Mississippi River Basin under the Renewable Fuel Standard.” *Environmental Science & Technology*. (IF: 9.028, second author) [LINK]
- Madhu Khanna, **Luoye Chen**, Bruno Basso, Ximing Cai, John Field, Kaiyu Guan, Chongya Jiang, Tyler Lark, Tom Richard, Seth Spawn-Lee, Peng Yang, 2021. “Redefining Marginal Land for Bioenergy Crop Production.” *GCB Bioenergy*. (IF: 4.745, second author) [LINK]
- Chongya Jiang, Kaiyu Guan, Madhu Khanna, **Luoye Chen**, Jian Peng, 2021. “Assessing Marginal Land Availability Based on Land Use Change Information in the Contiguous United States.” *Environmental Science & Technology*. (IF: 9.028, contribute author) [LINK]

- Shengxiao Li, **Luoye Chen**, Pengjun Zhao, 2019. “The Impact of Metro Services on Housing Prices: a Case Study from Beijing.” *Transportation*. (IF: 5.192, second author) [LINK]

SELECTED PRESENTATIONS

- “Weather Shocks and Economic Triggers of Cropland Change in the US: A Fine-scale Spatial Analysis”

American Geophysical Union (**AGU**) annual meeting (virtual), Nov 2020.

American Economic Association (**AEA**) annual meeting (virtual), Jan 2021.

Association of Environmental and Resource Economics (**AERE**) annual meeting (virtual), Jun 2021.

Agricultural and Applied Economics Association (**AAEA**) annual meeting, Austin, Aug 2021.

- “Fowl Climate: Assessing the Heterogeneous Impact of Climate Change on Bird Biodiversity in United States”

22nd international Biodiversity and Economics for Conservation Conference (**BIOECON XXII**), Wyoming, Sep 2021.

AERE annual meeting, Lake Tahoe, Jun 2019.

AAEA annual meeting, Poster Presentation, Washington DC, Aug 2018.

SERVICE

Reviewer: Research in Transportation Economics

TEACHING

ACE MS/MAAE Students Summer Math Workshop, UIUC

Instructor with full responsibility (Graduate module, 2021)

ACE 592: Data Science for Applied Economics, UIUC

Guest Speaker (Graduate module, 2021)

ACE 261: Applied Statistical Methods, UIUC

Teaching Assistant (Undergraduate module, 2020)

FELLOWSHIPS AND AWARDS

Outstanding Ph.D. Student (<i>ACE Department Award for top two graduate students</i>)	2021
ACES Fellowship	2016-2017
Peking University Outstanding Graduate Student	2016
Peking University Graduate Fellowship	2014-2015
China National Scholarship	2012

MISCELLANEOUS

- Computer Skills: Python, Stata, R, ArcGIS/QGIS, GAMS, $\text{\LaTeX}$
- Languages: English(fluent), Mandarin(native)
- Affiliations: Agricultural and Applied Economics Association (AAEA), Association of Environmental and Resource Economists (AERE), American Economic Association (AEA), American Geophysical Union (AGU).

REFERENCES

Madhu Khanna (ADVISOR)

ACES Distinguished Professor

Department of ACE

University of Illinois at Urbana-Champaign

husted@illinois.edu (Office Administrator: Amy Rosenberg)

Peter Christensen

Assistant Professor

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Weather Shocks and Economic Triggers of Cropland Change in the US: A Fine-scale Spatial Analysis (Job Market Paper)

Land conversion between cultivated cropland and grassland is a pressing concern because of its substantial implications for food supply and ecosystem services. However, there has been limited understanding of the frequency and determinants of the crop-grass conversion. Using high-resolution data on 500-meter land parcels in the rainfed United States from 2001 to 2017, we provide the first large-scale investigation of how US farmers adjust the crop-grass conversion in response to weather shocks and commodity price fluctuations. We also characterize heterogeneity in these adjustments across land parcels with different biophysical attributes using generic machine learning inference. We find that the contemporaneous rainfall shocks, long-term climate change, and lagged received commodity prices jointly affect the likelihood of crop-grass conversions. The effects of rainfall shocks and price fluctuations on crop-grass conversions are mainly confined to land parcels at the margin of agricultural production, with relatively lower water storage capacity, less productivity, and steeper slopes. These findings suggest that agricultural land use is not a static phenomenon and that it is important to examine how it evolves in response to various factors that may change over time.

Air Pollution and Flight Delays in China

Flight delays adversely affect millions of passengers worldwide every year, with substantial economic costs. Although a long list of factors are known to contribute to such delays, the effect of air pollution on flight delays has never been assessed. Here, we analyze how the concentration of PM_{2.5} affects flight departure delays using daily data from 131 airports in China, the world's largest aviation market. We find that, on average, a one-standard-deviation increase in the daily PM_{2.5} concentration leads to roughly 13 additional minutes in gross departure delay per flight and 3 additional minutes in net departure delay per flight after excluding arrival delays from previous assignments. These effects are more pronounced for larger airline companies and for larger airports. In turn, decreases in PM_{2.5} concentrations attributable to the Air Pollution Prevention and Control Action Plan in 2013 accounted for 15-33% of the total reduction in national average net flight delays in 2016 and 2017. In total, we estimate reductions in China's PM_{2.5} pollution from 2016 to 2017 avoided 2.4-3.8 billion US\$ in passenger time and labor losses, offsetting approximately 5-8% of China's average annual costs of PM_{2.5} control measures 2013-2017. Our results highlight a previously neglected impact of air pollution and further support China's ambitious clean air policies.